

MVP Prototype Design Specification

NeuroTriage AI v1.0

Purpose

This document defines the expected behaviour of the NeuroTriage AI MVP prototype. It complements the Product Requirements Document (PRD) by focusing on the implementation to be demonstrated as part of the academic project. The objective is to demonstrate how AI integrates into an existing radiology workflow rather than the performance of an AI model.

1. Prototype Objectives

- Simulate arrival of new CT Brain studies.
- Demonstrate AI-assisted worklist prioritization.
- Display a realistic PACS-inspired worklist.
- Display real CT brain images during interpretation.
- Present AI findings transparently.
- Capture radiologist confirmation or override.
- Illustrate the workflow from image acquisition to reporting.

2. Prototype Workflow

Initial Worklist → Simulate New CT Study → Study Status: Acquiring → Study Status: Completed + AI Status: Processing → AI Status: Ready → Priority Assigned → Study Moves to Top → Reading Screen → Confirm/Override → Study Status: Reported

3. Screen 1 – PACS Worklist

Columns: Patient Name, Study, Study Status, AI Status, AI Priority.

Study Status: Acquiring, Completed, Reported.

AI Status: Processing, Ready.

Clicking 'Simulate New CT Study' creates a new study that progresses through Acquiring → Completed → AI Processing → Ready before being reprioritized automatically.

4. Screen 2 – Reading Experience

Left: Real CT image viewer with slice navigation.

Right: AI Assessment panel showing Prediction, Confidence, Priority, Hemorrhage Location and AI Overlay toggle.

Bottom: Confirm and Override actions.

5. Feedback Behaviour

Confirm or Override records the decision, briefly displays 'Feedback Recorded', and returns to the worklist. Confirmed studies update their Study Status to Reported.

6. Sample Demonstration Data

The worklist initially contains 5–6 completed studies. A simulated incoming study progresses through the workflow and is reprioritized according to the AI policy.

7. Future AI Integration

The first implementation may use simulated AI outputs. The architecture should support replacing them with a real inference engine later without changing the UI.

8. Technical Goals

Demonstrate realistic clinical workflow, intuitive interaction, seamless AI integration, transparent recommendations, and minimal disruption to existing PACS workflow.

9. Demonstration Script

1. Open application.
2. Review worklist.
3. Click 'Simulate New CT Study'.
4. Observe acquisition and AI processing.
5. Observe automatic reprioritization.
6. Open the study.
7. Review CT images and AI overlay.
8. Confirm finding.
9. Return to worklist showing Study Status = Reported.